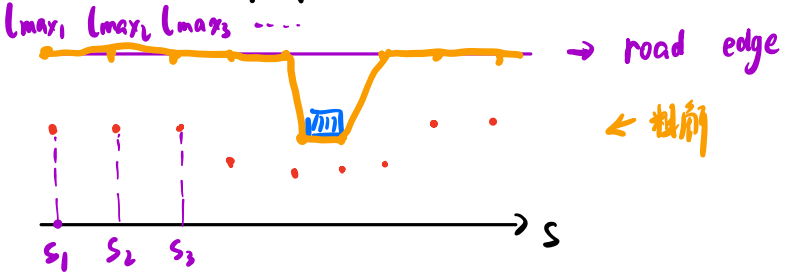
自动驾驶决策规划算法第二章第四节(中)

二次规划算法





l_{min1} l_{min2} ..... 与 l_{max1} l_{max2} ... 构成凸空间

二次规划的求解空间就在此凸空间中搜索

已知 s_1 $[l_{min1}, l_{max1}]$

s_2 $[l_{min2}, l_{max2}]$

s_n $[l_{minn}, l_{maxn}]$

求 $l = f(s)$ 满足: $f(s)$ 二阶导数连续

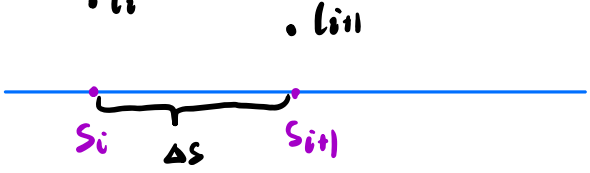
$l_{mini} \le f(s_i) \le l_{maxi}$ $i = 1, 2, 3, \cdots n$

$f(s)$ 尽可能平滑 ($f(s)$ 的各阶导数的平方尽可能小)

Apollo 3.5/5.0 新增, $f(s)$ 尽可能在凸空间的中央

分段加加速度优化法

记 $l_i = f(s_i)$ $l_i' = f'(s_i)$ $l_i'' = f''(s_i)$



假设连接 l_i 与 l_{i+1} 的曲线 $f(s)$ 的三阶导数恒为 $\frac{l_{i+1}'' - l_i''}{\Delta s}$ (常数)

在 l_i 与 l_{i+1} 之间 $f(s)$ 的四阶导数及以上皆为零

那么 l_{i+1}, l_{i+1}' 可以进行有限项的泰勒展开

$$l_{i+1} = l_i + l_i' \Delta s + \frac{1}{2} l_i'' \Delta s^2 + \frac{1}{6}\left(\frac{l_{i+1}'' - l_i''}{\Delta s}\right)\Delta s^3$$

$$l_{i+1}' = l_i' + l_i'' \Delta s + \frac{1}{2}\left(\frac{l_{i+1}'' - l_i''}{\Delta s}\right)\Delta s^2$$

$$\Rightarrow \quad l_i + \Delta s l_i' + \frac{1}{3}\Delta s^2 l_i'' - l_{i+1} + \frac{1}{6}\Delta s^2 l_{i+1}'' = 0$$

$$l_i' + \frac{1}{2}\Delta s l_i'' - l_{i+1}' + \frac{1}{2}\Delta s l_{i+1}'' = 0$$

$$\Rightarrow \begin{pmatrix} 1 & \Delta s & \frac{1}{3}\Delta s^2 & -1 & 0 & \frac{1}{6}\Delta s^2 \\ 0 & 1 & \frac{1}{2}\Delta s & 0 & -1 & \frac{1}{2}\Delta s \end{pmatrix} \begin{pmatrix} l_i \\ l_i' \\ l_i'' \\ l_{i+1} \\ l_{i+1}' \\ l_{i+1}'' \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

记 $A_{eq\text{-}sub} = \begin{pmatrix} 1 & \Delta s & \frac{\Delta s^2}{3} & -1 & 0 & \frac{\Delta s^2}{6} \\ 0 & 1 & \frac{\Delta s}{2} & 0 & -1 & \frac{\Delta s}{2} \end{pmatrix}$ 2×6

设有 $l_1, l_1' l_1'' \cdots l_n l_n' l_n''$ 需要优化

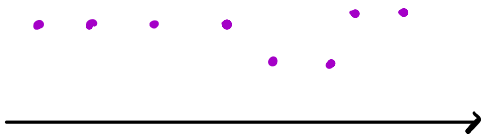
则分段加加速度约束为

$$\begin{pmatrix} A_{eq\text{-}sub} & 0\ 0\ 0\ \cdots & & \\ 0\ 0\ 0 & A_{eq\text{-}sub} & 0\ 0\ \cdots & \\ 0\ 0\ 0\ 0\ 0\ 0 & & A_{eq\text{-}sub} & \\ & & & \ddots \end{pmatrix} \begin{pmatrix} l_1 \\ l_1' \\ l_1'' \\ l_2 \\ l_2' \\ l_2'' \\ \vdots \\ l_n \\ l_n' \\ l_n'' \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

$2n-2 \times 3n$ $\quad 3n\times 1$ $\quad 2n-2\times 1$

记为 $A_{eq}\, x = b_{eq}$ 等式约束

不等式约束





$l_{min1} \le l_1 \le l_{max1}$

$$\begin{pmatrix} l_1 \\ l_1' \\ l_1'' \\ l_2 \end{pmatrix} \quad \begin{pmatrix} l_{min1} \\ -\infty \\ -\infty \\ l_{min2} \end{pmatrix} \quad \begin{pmatrix} l_{max1} \\ +\infty \\ +\infty \\ l_{max2} \end{pmatrix}$$

$l_{min2} \le l_2 \le l_{max2}$

⋮

$\Rightarrow$

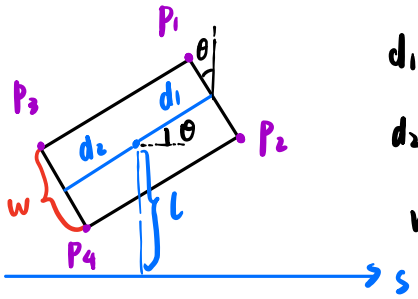
$x = \begin{pmatrix} l_1 \\ l_2 \\ \vdots \end{pmatrix} \quad \begin{pmatrix} -\infty \\ -\infty \\ \vdots \end{pmatrix} \le x \le \begin{pmatrix} +\infty \\ +\infty \\ \vdots \end{pmatrix}$

不准确

汽车不是质点，有体积



需要对汽车的四个角点做约束



d_1：汽车质心到车头距离

d_2 ⋯⋯ ⋯ 车尾 ⋯

w：汽车宽度

$l_{p1} = l + d_1 \sin\theta + \frac{w}{2}\cos\theta$

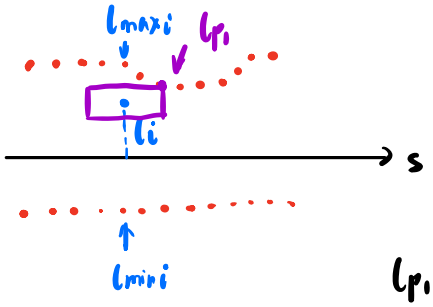
$l_{p2} = l + d_1 \sin\theta - \frac{w}{2}\cos\theta$

$l_{p3} = l - d_2 \sin\theta + \frac{w}{2}\cos\theta$

$l_{p4} = l - d_2 \sin\theta - \frac{w}{2}\cos\theta$

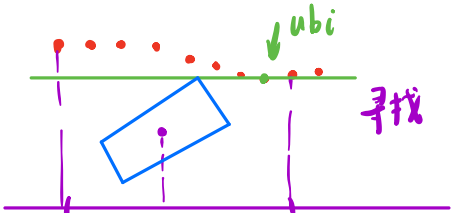
存在三角函数，非线性，不得不做一些近似处理

$\sin\theta \approx \tan\theta \approx l'$ $\quad \cos\theta \approx 1$



$l_{p1} = l_i + d_1 l_i' + \frac{w}{2} \in (l_{mini}, l_{maxi})$ 不太合适

做的保守一些



寻找 $(s_i - d_2, s_i + d_1)$ 中的 l_{maxi} 的最小值，记为 ub_i

$l_{p1} \le ub_i$

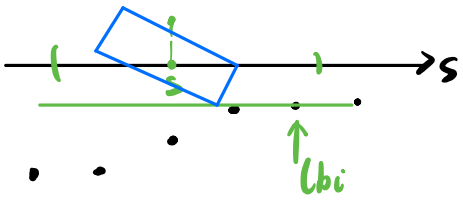
$s_i - d_2$　s_i　$s_i + d_1$

$l_{p_2} \le ub_i$

$l_{p_3} \le ub_i$

$l_{p_4} \le ub_i$

对 l_{min_i} 类似，找到 $(s_i - d_2, s_i + d_1)$ 中 l_{min} 的**最大值**，记为 lb_i



$l_{p_1} \ge lb_i$

$l_{p_2} \ge lb_i$

$l_{p_3} \ge lb_i$

$l_{p_4} \ge lb_i$

不等式约束

$$lb_i \le l_i + d_1 l_i' + \frac{w}{2} \le ub_i$$

$$lb_i \le l_i + d_1 l_i' - \frac{w}{2} \le ub_i$$

$$lb_i \le l_i - d_2 l_i' + \frac{w}{2} \le ub_i$$

$$lb_i \le l_i - d_2 l_i' - \frac{w}{2} \le ub_i$$

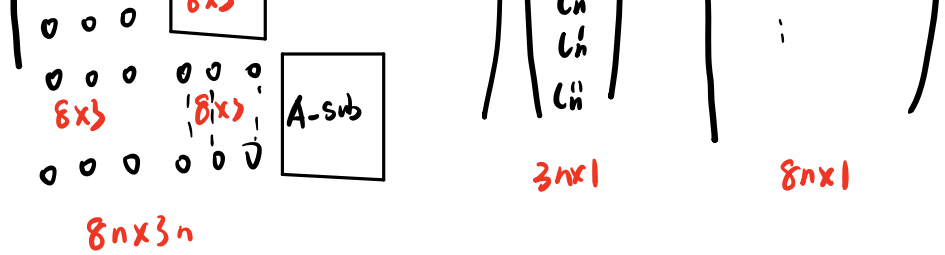
$$\begin{pmatrix} lb_i - \frac{w}{2} \\ lb_i + \frac{w}{2} \\ lb_i - \frac{w}{2} \\ lb_i + \frac{w}{2} \end{pmatrix} \le \begin{pmatrix} 1 & d_1 & 0 \\ 1 & d_1 & 0 \\ 1 & -d_2 & 0 \\ 1 & -d_2 & 0 \end{pmatrix} \begin{pmatrix} l_i \\ l_i' \\ l_i'' \end{pmatrix} \le \begin{pmatrix} ub_i - \frac{w}{2} \\ ub_i + \frac{w}{2} \\ ub_i - \frac{w}{2} \\ ub_i + \frac{w}{2} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & d_1 & 0 \\ 1 & d_1 & 0 \\ 1 & -d_2 & 0 \\ 1 & -d_2 & 0 \\ -1 & -d_1 & 0 \\ -1 & -d_1 & 0 \\ -1 & d_2 & 0 \\ -1 & d_2 & 0 \end{pmatrix} \begin{pmatrix} l_i \\ l_i' \\ l_i'' \end{pmatrix} \le \begin{pmatrix} ub_i - \frac{w}{2} \\ ub_i + \frac{w}{2} \\ ub_i - \frac{w}{2} \\ ub_i + \frac{w}{2} \\ -lb_i + \frac{w}{2} \\ -lb_i - \frac{w}{2} \\ -lb_i + \frac{w}{2} \\ -lb_i - \frac{w}{2} \end{pmatrix}$$

记为 $A_sub \begin{pmatrix} l_i \\ l_i' \\ l_i'' \end{pmatrix} \le b_sub_i$

总的不等式约束为

$$\begin{pmatrix} A_sub & 0 & 0 & \cdots \\ 8\times 3 & & & \\ 0 & A_sub & 0 & \cdots \\ 8\times 3 & 8\times 3 & & \\ 0 & 0 & A_sub & \cdots \\ 8\times 3 & 8\times 3 & 8\times 3 & \\ \vdots & \vdots & \vdots & \end{pmatrix} \begin{pmatrix} l_1 \\ l_1' \\ l_1'' \\ \vdots \\ l_n \\ l_n' \\ l_n'' \end{pmatrix} \le \begin{pmatrix} b_sub_1 \\ 8\times 1 \\ b_sub_2 \\ 8\times 1 \\ \vdots \end{pmatrix}$$



记为 $Ax \le b$　　$A\ (8n \times 3n)$　　$x\ (3n \times 1)$　　$b\ (8n \times 1)$

$$\text{cost function} = W_{cost_ref} \cdot \left(\sum_i l_i^2\right) + W_{cost_dl} \cdot \left(\sum_i l_i'\right)^2 + W_{cost_ddl} \cdot \left(\sum_i l_i''^2\right) + W_{cost_dddl} \cdot \left[\sum_i \left(l_{i+1}'' - l_i''\right)^2\right] + W_{cost_mid} \cdot \left[\sum_i \left(l_i - \frac{l_{min\,i} + l_{max\,i}}{2}\right)^2\right]$$

参考线代价　一阶导数代价　二阶导数代价　三阶导数代价　在凸空间中央代价

约束　$Aeq\ x = beq$　　$Ax \le b$

$$x = \begin{pmatrix} l_1 \\ l_1' \\ l_1'' \\ \vdots \\ l_n \\ l_n' \\ l_n'' \end{pmatrix}$$

求解出 $l_1\ l_1'\ l_1''\ \cdots\ l_n, l_n'\ l_n''$ 与 $s_1\ s_2\ \cdots\ s_n$ 结合即得到二次规划下的最优路径

在转化为 Cartesian 坐标，路径规划完成